

1-

```
school_db=# CREATE OR REPLACE FUNCTION multiply(num1 NUMERIC, num2 NUMERIC)
school_db=# RETURNS NUMERIC AS $$
school_db$# BEGIN
school_db$#     RETURN num1 * num2;
school_db$# END;
school_db$# $$ LANGUAGE plpgsql;
CREATE FUNCTION
school_db=# SELECT multiply(5, 7);
multiply
-----
        35
(1 row)
```

2-

```
school_db=# CREATE OR REPLACE FUNCTION hello_world(name VARCHAR)
school_db=# RETURNS VARCHAR AS $$
school_db$# BEGIN
school_db$#     RETURN 'Welcome, ' || name || '!';
school_db$# END;
school_db$# $$ LANGUAGE plpgsql;
CREATE FUNCTION
school_db=#
school_db=# SELECT hello_world('mennan');
hello_world
-----
Welcome, mennan!
(1 row)
```

3-

```

school_db=# CREATE OR REPLACE FUNCTION check_odd_even(num INTEGER)
school_db=# RETURNS VARCHAR AS $$
school_db$# BEGIN
school_db$#     IF num % 2 = 0 THEN
school_db$#         RETURN num || ' is even';
school_db$#     ELSE
school_db$#         RETURN num || ' is odd';
school_db$#     END IF;
school_db$# END;
school_db$# $$ LANGUAGE plpgsql;
CREATE FUNCTION
school_db=# SELECT check_odd_even(4);
  check_odd_even
-----
 4 is even
(1 row)

```

4-

```

school_db=# CREATE OR REPLACE FUNCTION get_student_info(student_id INTEGER)
school_db=# RETURNS TABLE (
school_db(#     id INTEGER,
school_db(#     e_name VARCHAR,
school_db(#     email VARCHAR,
school_db(#     address TEXT,
school_db(#     track_name VARCHAR,
school_db(#     birth_date DATE,
school_db(#     gender VARCHAR
school_db(# ) AS $$
school_db$# BEGIN
school_db$#     RETURN QUERY
school_db$#     SELECT s.id, s.e_name, s.email, s.address, t.track_name, s.birth_date, s.gender
school_db$#     FROM student s
school_db$#     LEFT JOIN track t ON s.track_id = t.id
school_db$#     WHERE s.id = student_id;
school_db$# END;
school_db$# $$ LANGUAGE plpgsql;
CREATE FUNCTION
school_db=# SELECT * FROM get_student_info(1);
 id | e_name | email | address | track_name | birth_date | gender
-----+-----+-----+-----+-----+-----+-----
  1 | John Doe | john@example.com | 123 Main St | Computer Science | 1990-05-15 | Male
(1 row)

```

5-

```

school_db=# CREATE OR REPLACE FUNCTION get_subject_avg(subject_name VARCHAR)
school_db=# RETURNS NUMERIC AS $$
school_db=# DECLARE
school_db=#     avg_grade NUMERIC;
school_db=# BEGIN
school_db=#     SELECT AVG(g.grade) INTO avg_grade
school_db=#     FROM grades g
school_db=#     JOIN subject s ON g.sub_id = s.id
school_db=#     WHERE s.sub_name = subject_name;
school_db=#
school_db=#     RETURN ROUND(avg_grade, 2);
school_db=# END;
school_db=# $$ LANGUAGE plpgsql;
CREATE FUNCTION
school_db=# SELECT get_subject_avg('Mathematics');
  get_subject_avg
-----
      89.25
(1 row)

```

6-

```

school_db=# -- 1. First create the archive table
school_db=# CREATE TABLE deleted_students (
school_db=#     id INTEGER,
school_db=#     e_name VARCHAR(100),
school_db=#     email VARCHAR(100),
school_db=#     address TEXT,
school_db=#     track_id INTEGER,
school_db=#     birth_date DATE,
school_db=#     gender VARCHAR(10),
school_db=#     deletion_time TIMESTAMP DEFAULT NOW()
school_db=# );
CREATE TABLE
school_db=#
school_db=# -- 2. Create the trigger function
school_db=# CREATE OR REPLACE FUNCTION archive_deleted_student()
school_db=# RETURNS TRIGGER AS $$
school_db=# BEGIN
school_db=#     INSERT INTO deleted_students (id, e_name, email, address, track_id, birth_date, gender)
school_db=#     VALUES (OLD.id, OLD.e_name, OLD.email, OLD.address, OLD.track_id, OLD.birth_date, OLD.gender);
school_db=#     RETURN OLD;
school_db=# END;
school_db=# $$ LANGUAGE plpgsql;
CREATE FUNCTION
school_db=#
school_db=# -- 3. Create the trigger
school_db=# CREATE TRIGGER before_student_delete
school_db=# BEFORE DELETE ON student
school_db=# FOR EACH ROW
school_db=# EXECUTE FUNCTION archive_deleted_student();
; CREATE TRIGGER

```

7-

```

school_db=# CREATE TABLE student_audit_log (
school_db=#     action_id SERIAL PRIMARY KEY,
school_db=#     student_id INTEGER,
school_db=#     action_type VARCHAR(10),
school_db=#     action_time TIMESTAMP DEFAULT NOW(),
school_db=#     old_data JSONB,
school_db=#     new_data JSONB,
school_db=#     description TEXT
school_db=# );
CREATE TABLE
school_db=#
school_db=# -- 2. Create the trigger function
school_db=# CREATE OR REPLACE FUNCTION log_student_changes()
school_db=# RETURNS TRIGGER AS $$
school_db=# DECLARE
school_db=#     action_desc TEXT;
school_db=# BEGIN
school_db=#     IF TG_OP = 'INSERT' THEN
school_db=#         action_desc := 'New student added: ' || NEW.e_name;
school_db=#         INSERT INTO student_audit_log (student_id, action_type, new_data, description)
school_db=#         VALUES (NEW.id, 'INSERT', to_jsonb(NEW), action_desc);
school_db=#     ELSIF TG_OP = 'UPDATE' THEN
school_db=#         action_desc := 'Student updated: ' || NEW.e_name;
school_db=#         INSERT INTO student_audit_log (student_id, action_type, old_data, new_data, description)
school_db=#         VALUES (NEW.id, 'UPDATE', to_jsonb(OLD), to_jsonb(NEW), action_desc);
school_db=#     ELSIF TG_OP = 'DELETE' THEN
school_db=#         action_desc := 'Student deleted: ' || OLD.e_name;
school_db=#         INSERT INTO student_audit_log (student_id, action_type, old_data, description)
school_db=#         VALUES (OLD.id, 'DELETE', to_jsonb(OLD), action_desc);
school_db=#     END IF;
school_db=#     RETURN NULL;
school_db=# END;
school_db=# $$ LANGUAGE plpgsql;
CREATE FUNCTION
school_db=#
school_db=# -- 3. Create the trigger
school_db=# CREATE TRIGGER after_student_changes
school_db=# AFTER INSERT OR UPDATE OR DELETE ON student
school_db=# FOR EACH ROW
school_db=# EXECUTE FUNCTION log_student_changes();
CREATE TRIGGER

```

testing:

```

school_db=# INSERT INTO student (e_name, email, address, track_id, birth_date, gender)
school_db=# VALUES ('Test Student', 'test@example.com', '123 Test St', 1, '2000-01-01', 'Male');
INSERT 0 1
school_db=#
school_db=# SELECT * FROM student_audit_log WHERE action_type = 'INSERT';
 action_id | student_id | action_type |      action_time      | old_data | new_data | description
-----+-----+-----+-----+-----+-----+-----
1 | 9 | INSERT | 2025-05-01 13:52:43.481503 | {"id": 9, "email": "test@example.com", "e_name": "Test Student", "gender": "Male", "address": "123 Test St", "track_id": 1, "birth_date": "2000-01-01"} | {"id": 9, "email": "test@example.com", "e_name": "Test Student", "gender": "Male", "address": "123 Test St", "track_id": 1, "birth_date": "2000-01-01"} | New student added: Test Student
(1 row)

school_db=# UPDATE student SET email = 'updated@example.com' WHERE e_name = 'Test Student';
UPDATE 1
school_db=# SELECT * FROM student_audit_log WHERE action_type = 'UPDATE';
 action_id | student_id | action_type |      action_time      | old_data | new_data | description
-----+-----+-----+-----+-----+-----+-----
2 | 9 | UPDATE | 2025-05-01 13:53:11.035104 | {"id": 9, "email": "test@example.com", "e_name": "Test Student", "gender": "Male", "address": "123 Test St", "track_id": 1, "birth_date": "2000-01-01"} | {"id": 9, "email": "updated@example.com", "e_name": "Test Student", "gender": "Male", "address": "123 Test St", "track_id": 1, "birth_date": "2000-01-01"} | Student updated: Test Student
(1 row)

school_db=# DELETE FROM student WHERE e_name = 'Test Student';
DELETE 1
school_db=# SELECT * FROM student_audit_log WHERE action_type = 'DELETE';
 action_id | student_id | action_type |      action_time      | old_data | new_data | description
-----+-----+-----+-----+-----+-----+-----
3 | 9 | DELETE | 2025-05-01 13:53:39.197423 | {"id": 9, "email": "updated@example.com", "e_name": "Test Student", "gender": "Male", "address": "123 Test St", "track_id": 1, "birth_date": "2000-01-01"} | {"id": 9, "email": "updated@example.com", "e_name": "Test Student", "gender": "Male", "address": "123 Test St", "track_id": 1, "birth_date": "2000-01-01"} | Student deleted: Test Student
(1 row)

```

```
school_db=# SELECT
school_db=#         action_id,
school_db=#         student_id,
school_db=#         action_type,
school_db=#         action_time,
school_db=#         description
school_db=# FROM student_audit_log
school_db=# ORDER BY action_time DESC;
 action_id | student_id | action_type |          action_time          | description
-----+-----+-----+-----+-----
          3 |          9 | DELETE     | 2025-05-01 13:53:39.197423 | Student deleted: Test Student
          2 |          9 | UPDATE     | 2025-05-01 13:53:11.035104 | Student updated: Test Student
          1 |          9 | INSERT     | 2025-05-01 13:52:43.481503 | New student added: Test Student
(3 rows)
```